



Review on recent research and technical challenges of floatover installation operation

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ABSTRACT

The floatover installation method is one of the two installation methods of the offshore platform topsides and is a much safe and efficient method for the mega integrated topsides. This paper presents a literature review on the research and technical challenges of floatover installations. Various floatover installation methods and concepts for the fixed and floating substructures with passive and active load transfer systems are reviewed, including the conventional single barge floatover installation, the T-barge floatover installation, the twin/dual floater floatover installation, and the dynamically positioning floatover installation. The exclusive mating units including the deck/leg mating units and the related guidelines are discussed. Investigations in respect of the in-site measurements, numerical modeling, and physical model tests are presented, following challenges and gaps between the practices and the research. Finally, further developing and refining modeling techniques are envisioned. This review aims to provide suggestions on research and development activities of floatover installation.

1. Introduction

The topsides installation of the offshore platform is one of the expensive and important tasks. The topsides installation methods include the lifting method and the floatover method. In the lifting method, the individual module or the integrated topsides are lifted by the floating crane vessel and mated with the substructures. Lifting the individual functional module separately and welding modules together are time-consuming and uneconomical. Lifting the integrated topsides installation requires a large floating crane and an extended crane radius. Those factors result in a great limitation of the lifting operations and constraint the application. The floatover method has great advantages over the lifting method when it comes to the mega topsides exceeding the capacity of the lifting crane. The floatover method allows that more than 20000-ton mega topsides are fabricated and outfitted onshore. The integrated topsides are then placed on the barge deck in one piece and finally transferred onto the substructures by the tide, the ballasting system, or the active jacking system. Besides the capacity, there are plenty of benefits and strengths of the integrated topside floatover method. More flexibility of the barge choice, less long-term offshore hook-up operations, and reductions of schedule interfaces make the

floatover method a much safer and more cost-efficient method than the lifting method (Abbot et al., 1980; Gros and Lescurat, 1982; Ramzan and Rawstron, 1984; White et al., 1986; Beal et al., 1992; Seij and Groot, 2007; Kocaman et al., 2008; Cholley et al., 2009; Wang et al., 2010; Liu and Li, 2017).

The concept of the topsides floatover method is introduced as early as the late 1970s and attracted the great attention of engineers and researchers in the past decades. Numerous floatover techniques with different load transfer systems and barge configurations are developed to adapt different topsides and substructure sizes as well as various sea state conditions. The weight of the installed topsides increases from about 6000 tons in Zakum field, Abu Dhabi to 37,000 tons in Hibernia field, Canada. The floatover method is successfully utilized in the sheltered water and the open water in Africa, Asia, Australia, the Middle East, Canada, etc (Kenney et al., 1994; Labbe and Legras, 1998; O'Neill et al., 2000; Maher et al., 2001; Cholley and Cahay, 2007; Tribout et al., 2007; Mouhandiz and Bokhorst, 2013; Wu et al., 2014; Joshi, 2018). The floatover method also provides a potential solution for offshore wind turbine installations (Cholley and Cahay, 2009; Chen et al., 2020), the offshore electric substation installation (<https://www.ctgne.com/sxxnychina/1077761/1077661/1184423/index.html>), and the bridge installation (<https://www.crbc.com/site/crbcEN/381/info/2021/4688>

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Abbreviation list

LMU	Leg mating unit
DSU	Deck supporting unit
DSF	Deck support frame
DTV	Deck transportation vessel
DP	Dynamic positioning
CoG	Center of gravity
GBS	Gravity based substructure
TLP	Tension leg platform
DOE	Design of experiment
INS	Inertial navigation system
GPS	Global positioning system
RTK	Real time kinematic
ADCP	Acoustic Doppler current profilers

3920.html). In China, 54 floatover projects for the fixed platform have been carried out in the Bohai Sea, the East China Sea, and the South China Sea. Table 1 lists the main projects in China. The maximum weight of the topsides reaches 33,000 tons in LW 3–1 project in the South China Sea. The installation operations are carried out in both shallow water and deep water. The maximum water depth of the floatover operation in

Table 1
Main floatover projects in China.

Project name	Location	Vessel	Data	Topside weight
BZ19-4	Bohai Bay	HYSY228, T-barge	August 2021	12,500 ton
CFD6-4	Bohai Bay	HYSY228, T-barge	January 2021	12,500 ton
CFD11-6	Bohai Bay	HYSY228, T-barge	August 2019	13,000 ton
CFD11-1	Bohai Bay	HYSY228, T-barge	July 2019	13,500 ton
DF13-2	South China Sea	HYSY229, T-barge	April 2019	17,000 ton
BZ34-9	Bohai Bay	HYSY228, T-barge	January 2019	13,500 ton
PL13-9	Bohai Bay	HYSY228, T-barge	October 2018	14,000 ton
WC9-2	South China Sea	HYSY228, T-barge	May 2017	14,000 ton
BZ34-2	Bohai Bay	HYSY228, T-barge	May 2015	15,200 ton
KL10-1	Bohai Bay	HYSY228, T-barge	January 2015	12,400 ton
JZ25-1S	Bohai Bay	HYSY228, T-barge	January 2015	14,400 ton
JZ9-3	Bohai Bay	HYSY228, T-barge	September 2014	10,800 ton
HZ25-8	South China Sea	HYSY278, DP barge	May 2014	15,500 ton
QHD32-6CEPJ	Bohai Bay	HYSY228, T-barge	July 2014	14,400 ton
QHD32-6CEPI	Bohai Bay	HYSY228, T-barge	June 2014	14,000 ton
KL3-2	Bohai Bay	HYSY228, T-barge	January 2014	11,900 ton
BZ35-2	Bohai Bay	HYSY228, T-barge	October 2013	12,200 ton
LF7-2	South China Sea	HYSY221	August 2013	12,500 ton
LW3-1	South China Sea	HYSY229, T-barge	December 2012	33,000 ton
JZ1-1	Bohai Bay	HYSY221	August 2010	10,100 ton
LD27-2	Bohai Bay	HYSY221	August 2009	10,000 ton
JZ25-1	Bohai Bay	HYSY221	May 2009	5650 ton
BZ34-1	Bohai Bay	HYSY221	October 2007	8585 ton
NB35-2	Bohai Bay	HYSY221	May 2005	7200 ton

China is about 300 m. Although a lot of the platform topsides have been installed by the floatover method, the floatover method still faces great challenges, such as the harsher offshore environment, the limited weather windows, the deeper water, the growing topside weight, and the limited choice of the floatover barge. The floatover technique needs to be further improved.

It is worth mentioning that up to now there is few reviews reported to the public related to the recent research and developments on the floatover installations. Therefore, the objective of this study is to present a literature review on the research and development of floatover operations, as well as to highlight the analyzed methods and associated challenges. Section 2 reviews various floatover methods, the related standards and guidelines, and the exclusive equipment of the floatover operations. The following three sections review the in-site measurement, the experimental modeling, and the numerical modeling, respectively. The related critical technique problems and analysis methods are summarized, together with the hydrodynamic and structural response characteristics of different floatover systems. Section 5 presents the main conclusions and recommendations for future research.

2. Overview of topside floatover installation

2.1. Floatover installation method

The floatover installation generally includes several consecutive stages. 1) Load-out stage. After the integrated topsides are constructed in the quay, the topsides are placed onto the deck of the transportation barges, which is carried out with the assistance of the skidway and the tugs. 2) Transportation stage. The topsides are transported by the self-propulsion or the tug to the vicinity of the host structures/substructures by the transportation barge. 3) Pre-mating stage. The barge is positioned near the substructures by the mooring lines and is attached to the substructures by the lash lines. The fenders are prepared to assist the load transfer operations. 4) Mating stage. Once the weather window is available, the barge approaches into the slots of the substructures with help of the tug and the lash line winch. The DSUs are employed to support the topsides and to absorb the impact loads. After the stabbing pins on the topsides LMUs are aligned with the receptacles installed on the substructures, the topsides are transferred onto the substructures by the tide, the ballast system, or the active jacking system. 5) Post-mating stage. After all topsides weights are transferred onto the substructures, the barge is continuously ballasted down for enough clearances between pins and receptacles of DSUs and then towed away. Fig. 1 illustrates the general procedure of the floatover installation.

The floatover installation method can be classified based on different criteria. Based on the employed barge, the floatover method includes the single barge floatover method, the catamaran/twin-barge floatover method, and the dynamical positioning barge floatover method. If the criterion is the load transfer system, the floatover method includes the active jacking floatover method and the passive floatover method. The floatover installation method can be for the fixed structure floatover method or the floating structure floatover method if the criterion is the type of the substructure. Table 2 summarizes the classifications based on different criteria.

2.1.1. Classification based on barge

The topsides can be transported and installed by various barges, such as the single barge, twin barges/floaters, and the dynamically positioning barge. The single hull with the same width of the bow and the stern is the conventional installation barge for the topsides floatover operations, as shown in Fig. 2. In general, the topsides are placed close to the longitudinal center of the hull. Such configurations benefit the ballasting/de-ballasting operations during the load transfer and provide the stability of the floatover barge system. Various platform topsides have been successfully installed by the hull. Sigrist et al. (1998) applied the single barge for the heavy topsides in Arabian Gulf and Malaysia and

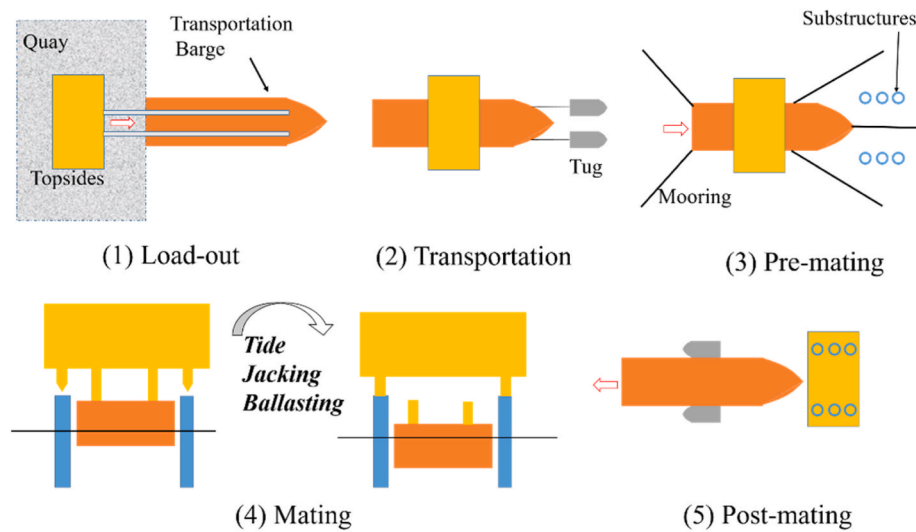


Fig. 1. The general procedure of the floatover operations.

Table 2
Classification based on different criteria.

Criterion	Classification	Pros	Cons
Barge	Barge	More choice of the barge	Limit capacity
	T-barge	Accommodate the slots of the substructure	Specific construction; Trim control
	Catamaran/twin-barge	Reuse the barges; Slots free; Fender free	Complex interactions; Difficulty load-out on barges;
	DP barge	Mooring free; Efficient for deep water	Expensive DP system
Transfer system	Passive method	Easy construction; Mature technology	High air gap; Slow; Limited weather window
	Active method	Fast controllable operation; Low air gap	Expensive jacking system
Substructure	GBS	Low price	Difficulty substructures installation
	Jacket spar	Easy installation; Adapt to the deep water	Limited slot span; Lack of slots; Complex interactions
	TLP	Adapt to the deep water	Complex interactions
	Semi-submersible	Adapt to the deep water	Complex interactions

presented associated requirements in terms of the jacking system, ballasting system, and shock absorbers. They addressed that the transportation and installation phases imposed key constraints on the determinations of the minimum size of the barge and the opening slot of the jacket structure to allow entry of the barge. Tribout et al. (2007) gave an overview of the so-called UNIDECK technology, in which ballasting and jacking systems were combined to improve the stability of the heavy transport vessel during the transportation phase and to avoid high dynamic impact during the installation phase. Xia et al. (2005) used the single hull to install the minimal platforms with the cantilevered method, in which the topsides are supported on steel trusses extending over the barge bow. The proposed configuration enabled the single barge to approach the jacket head-on and therefore has few limitations on the platform design. Duquesnay et al. (2013) addressed the two critical stages of the single barge floatover installation system, docking and undocking stages, by numerical modeling. The effect of the weather



Fig. 2. The floatover barge (<https://www.cnoocengineering.com/col/col29541/index.html>).

criteria, the mating units, the mooring/cross lines on the workability were presented. Zhu et al. (2021) analyzed the nonlinear motions and loads of the single barge with emphasis on investigating the interactions between the topsides and the barge. Generally speaking, the single hull floatover barge is the most widely used in floatover operations, especially for small and medium-size topsides. However, the limited slots of the substructure and the carrying capacity of the hull restrict the applications of the straight hull. For example, the width of the jackets slot is much smaller than the hull width.

The T-barge is designed to accommodate the slots of the substructures and to adopt the mega topsides. The T-barge has sponsons in the aft body and has different breadth at the bow and the stern. The narrow bow and the wide stern make it applicable for the limited slots of the substructures. Fig. 3 illustrates the T-barge. Based on the configurations of the substructures, the topsides can be supported and installed by the T-barge either at the stern or the bow. The unique configuration of the T-barge changes the locations of the topsides on the deck of the barge from the midship position to the barge bow, thus leading to different characteristics during the mating operations. In Bokhorst et al. (2011), the launch barge H-851 of Heerema Marine Contractors was converted into a T-shaped float-over barge with a relatively narrow bow section of 42 m and a width of 63 m able to install ultra heavy topsides with weights up to 38,000 tons. Cherian et al. (2014) discussed the effect of the criticality of the CoG in the transverse and vertical directions for the T-barges. Yang et al. (2014) applied the T-barge for the floatover installation at a water depth of 190 m in the South China Sea, in which



Fig. 3. T-barge.

the 26200-ton topsides were located at the bow of the T-barge. As the varying mass distribution of the barge affects the motions behaviors, understanding of the barge mass and hydrodynamic characteristics are required to determine the most suitable location of the deck on the barge. Yuan et al. (2014) numerically investigated the hydrodynamic response of the T-barge, demonstrating the variations of the trim angle of the T-barge with the increase of the transferred loads. Bai et al. (2020a, c) experimentally simulated the T-barge with the rapid load transfer technique and found that the trim angle of the barge varied significantly with the percentage of the transferred loads. Bai et al. (2020b) further investigated the loads on mating units during the continuous load transfer process for the T-barge floatover system. Although the T-barge alleviates the constraints of the slot of the substructures and broadens the choice of the available barge for the mega topsides, the T-shaped configurations require a more excellent ballasting system to increase the stability of the floatover system and to avoid the possible impact at the mating phases.

Another attempt to eliminate the need for a slot or “float through” for the substructures is the twin barges/floaters or catamaran floatover method, as shown in Fig. 4. The twin barge floatover method with large gaps allows the straddling of the platform and avoids the impacts on the substructures, thus benefiting the design of the platform structure. It can

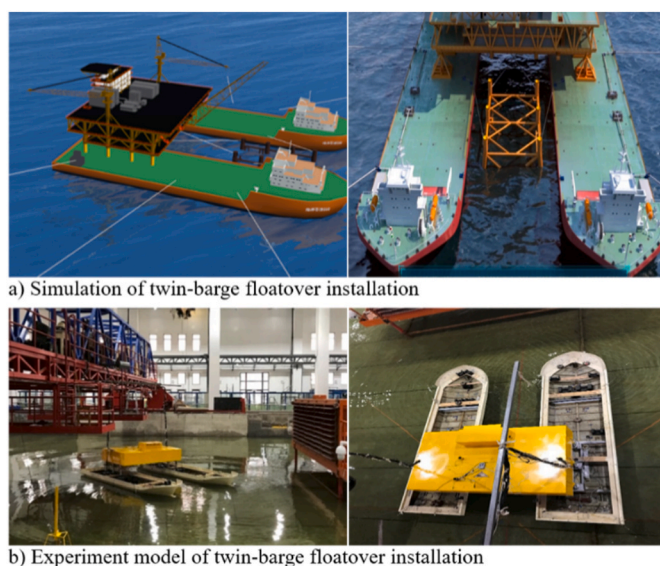


Fig. 4. Twin-barge floatover method.

also employ two relatively small barges/floaters to install a heaving deck, which allows to reuse existing barges and thus is economically efficient. The method is also a promising alternative for the topside installation of the spar and wind turbine platform. Tahar et al. (2006) developed the concept of the catamaran/twin-barge floatover method, in which two specially designed pontoon barges constituted a catamaran configuration. They carried out experimental and numerical investigations and demonstrated the favorable sea-keeping behavior in moderate sea states. Similarly, Edelson et al. (2008) successfully applied the twin-barge floatover installation technology for the 4000-ton deck installation of the Murphy Kikeh spar platform, which is the first application of the catamaran floatover installation in open water. Although the weight is relatively small, the successful execution proved the twin-barge method a viable and cost-effective alternative. They further presented the improved process of the installation system for heavy topsides having weights of up to 25,000 tons (Luo et al., 2013). The twin-barge floatover installation is further applied for the topsides of the truss spar FPSO in Norwegian (Vernes et al., 2019). The 25300-ton mega topsides were transferred onto the spar hull by the twin-barge, simultaneously ballasting of the dual barge hulls and de-ballasting of the spar hull. Magee et al. (2014) experimentally simulated the fork-like T-barge for the 2700-ton topsides installations in which the topsides were located at the fork-like stern.

Due to the twin barges/floaters configurations, the hydrodynamic and structural behavior of the topside-twin barge system is complex. Koo et al. (2010a, 2010b) experimentally and numerically simulated the twin barge floatover system and pointed out the stage of the synchronized motions between the topside and the twin-barge. They predicated the barge and substructure motions, the mooring force, fender force, and the mating loads at transportation, pre-mating, and load transfer stages and demonstrated the feasibility of performing the 17000-ton topsides floatover installation in the Gulf of Mexico. The structural strength of the topsides, the barges, and the connections should be investigated, especially for the quarter and beam seas. The strength of the topsides and barges system should be carefully examined. Luo et al. (2013) carried out the structural analysis of the catamaran system under static, survival, and operational load cases and estimated the maximum force and moments. Geba et al. (2017) used the quasi-static analysis to investigate the structural performance of the deck and presented the displacement of the topside exposed to the sagging moments. Edelson et al. (2008) conservatively checked the fatigue damage requirement of the sea fastening connections of the catamaran system although only two-week durations of the catamaran floatover operations were scheduled. Dessi et al. (2017a) investigated the sea-keeping characteristics of the rigid-connected twin-barge at the transportation phase by physical model tests.

In those practical applications, the twin barges were rigidly connected with the topsides, which would be optimized by the hinge connections to allow the relative roll rotations of the barge and topside and thus reduce the loads. To this end, Dessi et al. (2017b, 2019) investigated the elastically-connected twin-barge for the mega topside floatover installations with considerations of the topside stiffness, in which the seakeeping characteristics of the twin-barge were experimentally analyzed. They described the response difference between the single barge and the double barge transportation system. Bai et al. (2021a) utilized the twin-barge connected by the rotatable hemisphere mating units for a mega topsides installation, in which the roll motions of the barges were released. The loads on the mating units, the unsynchronized motions of the topsides and twin-barge, and the tether and mooring lines in various wave conditions were investigated by the physical model tests, together with the disturbed fluids in the gaps as shown in Fig. 5. However, the hydrodynamic and mechanical interactions of the multiple bodies for the twin-barge floatover system are still unclear and the available numerical tools for capturing the nonlinear coupling characteristics are few. Another challenge is the complex multi-body structural dynamics. For the twin-barge, the floatover system suffers from the

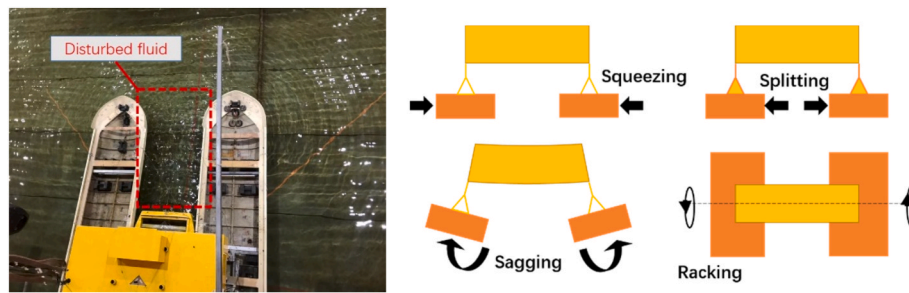


Fig. 5. Complex hydrodynamic and structural response.

sagging, squeezing, splitting, and racking loads in the quarter and beaming seas, as shown in Fig. 5. Geba et al. (2017) numerically demonstrated the sagging phenomenon based on the quasi-static approach and calculated the stress on the truss of the topsides. Edelson et al. (2008) evaluated the racking moment of the rigid-connected twin-barge and pointed out that there was the higher racking moment occurs in quartering seas. Due to the large waterplane area of the twin barges, racking moments and sagging bending moments governed the strength for the topside, the grillage, and the barges. It can be seen that the configurations of the twin-barge floatover system give rise to complex hydrodynamic and structural response, which is required to be investigated with considerations of the realistic configurations and environmental loads.

Load-out operation onto the twin-barge is one of the most critical steps and is much more difficult compared with the load-out operation onto a single barge. Fig. 6 illustrates the load-out operations from the twin-barge onto the single barge. For the topside load-out operations onto the single barge, either the shore-mounted cranes or the skidded method together with the trailer can be employed. For the mega topside load-out operations onto twin-barge, both mentioned methods are challenging, especially the separated twin barges. Edelson et al. (2008) presented the topside load-out operations from the transportation barge onto the twin-barge in the Kikeh projects, in which the topsides mounted on the transportation barge were connected with the twin barge one by one and then towed into the site of the substructures, although the operations lasted about one day and were carried out in the harbor. Xu et al. (2016) proposed the topsides load-out operations from the transportation barge onto the twin barge by ballasting the transportation barge and experimentally analyzed the hydrodynamic of the rigid-connected three barges. Bai et al. (2021b) proposed the topsides load-out operations from the T-shaped transportation barge onto the elastically-connected twin-barge at the same time, in which various load transfer stages of the three barges in different wave directions were experimentally investigated. Those investigations demonstrated that the load-out operations of the topsides onto the twin-barge are of great difficulty and are subjected to complex hydrodynamic and mechanical interactions.

The DP barge with excellent maneuverability and positioning

capability can eliminate the mooring and lash lines as well as the positioning tugs. Compared with the conventional floatover method assisted by complex moorings and lash lines, the dynamically positioning floatover method omits the preparation work at the pre-mating and post-mating phases, thus reducing the operational times (Beerendonk et al., 2008). The available short schedule makes the DP floatover method became an attractive alternative, especially for weather-sensitive operations. Several DP float-over installation projects have been performed in the last decades since the first practical installation was executed for the 9000-ton topsides in the Gulf of Thailand in 2003 using the DP-2 semisubmersible vessels, with multiple more in planning for execution in the next few years. Bakar (2019) presents the conversions from the deck transportation vessel to the DP-2 vessel, in which two portable were attached to the stern of the vessel. The 15500-ton mega topsides were successfully installed by the converted Portable DP Vessel in Sarawak, Malaysia within the South China Sea. The 20500-ton topsides of the Keababangan Central Processing Platform were installed in Malaysia in June 2014 and set the record of the heaviest platform installed by the DP floatover method (Zu et al., 2016). Bai et al. (2014), Xu et al. (2018), and Jin et al. (2018) presented comprehensive descriptions of the DP floatover technology and evaluated the feasibility in the East and the South China Sea by numerical simulations, model tests, virtual simulations, and field monitoring measurement. Lin et al. (2016) detailed the challenges of the topside installation in West Africa by the DP floatover method and recommended further improvements on the present DP floatover system to account for the challenging environmental condition. With the development of the dynamically positioning system, the DP floatover method extends its applications and plays an important role in the floatover installations in the harsh environment, such as the internal-wave-current-dominant South China Sea and swell-dominant West Africa. It should be noted that the DP system is designed to damp out the low-frequency motions in the horizontal plane and thus reduce the drift motions. While the wave-frequency motions of the vessel remain. The auxiliary fender system and LMU/DSU are still required to assist the positioning of the vessel. Li et al. (2015) and Yi et al. (2018) numerically simulated the DP floatover installation with the considerations of the sway and surge fenders and discussed the fender forces. The DP barge “HYSY278” and the configurations of the

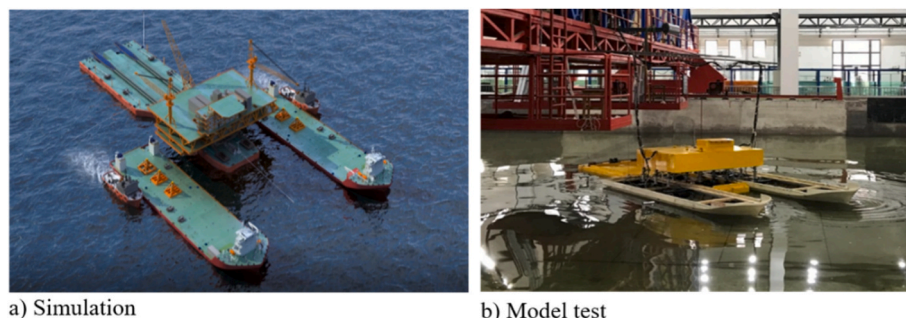


Fig. 6. Load-out operations onto twin-barge by the T-barge.

thrusters and propellers are shown in Fig. 7. He et al. (2020) proposed a time-domain simulation program to numerically analyze the fender forces as well as motions of the DP vessel in the floatover installation exposed to winds, waves, and currents. The corresponding physical model tests were conducted to validate the numerical model.

2.1.2. Classification based on the transfer system

The topsides are transferred on the substructures by either the non-mechanical method or the mechanical method, i.e., the passive method and the active method. Fig. 8 illustrates the passive method and the active method. The passive method, namely the non-mechanical method, is that the topsides weights are transferred onto the substructures by the ballasting system and the tide. The topsides and the barge move down together and keep the constant gaps between the barge deck and the topside. For the active method, the topsides are lift up and down by the mechanical system, such as the hydraulic jacks. The load transfer of the topsides is mainly by the mechanical system, in some cases assisted with the ballasting and the tide. The passive method needs a large elevation of the topsides before mating so that the topsides can be across the substructure, which poses difficulty for the transportation and the fabrication. In contrast, the active method allows the relatively small gap between the barge deck and topside before mating and benefits the transportations. Moreover, the quick and adjustable load transfer operations are available by the jacking system although the jacking system is expensive. There are also fewer weather limitations for the active method.

The HIDECK technology is a non-mechanical method and was firstly applied in 1983 (Mercier et al., 1991). The topsides were transported by the high DSF on a single barge and transferred onto substructures by increasing the draft of the barge together with the tide, as shown in Fig. 9. The installed topsides of the platform are generally required to be elevated enough gaps allowing the passage of design waves. The main feature of the HIDECK method is that the topsides were supported at a sufficient height at the transportation phase and were directly transferred onto the substructures without any active mechanical components for the elevation of the topsides. The HIDECK method is therefore classified as a passive floatover method. Although the concept initially was developed for the fixed platform, the method has been widely used for floaters such as TLP, semi-submersibles in the open water. The gap reduction can be improved not only by ballasting down the barge but by de-ballasting the floating substructure or the combinations. The largest challenge to HIDECK technology is the large air gap requirement for some platforms. The high elevation of the topsides requests the large DSF and leads to the low stability of the topsides-barge system, posing great difficulties for the transportation and mating stages.

The UNIDECK technology is an active mechanical floatover method (Tribout et al., 2007). The specially designed jacking system is employed to make the load transfer process controllable and reversible. Fig. 10 illustrates the floatover operations by the UNIDECK technology. The method allows that the topsides are transported at a relatively low elevation and then jacked up near the substructure before the docking operations. The load transfer process is accomplished by the rapid

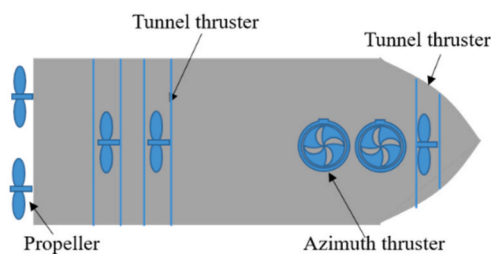
jacking down operations through the hydraulic jacks instead of relative slow ballasting operations, although the ballasting operations of the barge are in some cases needed when the strokes of the jacks are limited. The UNIDECK technology helps improve the stability of the topsides-barge system and is much safer because of the reversible operations. The use of the equipped jacking system makes this method capable of operating in severe swell conditions. By applying the jacking system, the UNIDECK technique enables short-time installation duration and ensures a safe installation operation, avoiding the high-frequency impact at the mating units at the initial and final mating stages.

The strand jack lifting technology is an active floatover method and was firstly applied in 2000. The strand jack technology raises the topside up by the strand jacks after the topsides are installed over the substructure legs, as illustrated in Fig. 11. The method allows that the topsides were transported at a very low level on the deck of the barge, thus increasing the barge stability and reducing the barge size. During the mating phase, the topsides support legs are firstly lowered to make initial contact with the pre-installed piles. This requires less or no ballasting for deck mating, eliminates the rapid ballast system. Another advantage is that the method significantly reduces the initial contact impact, just in the magnitude of the weight of the topside support leg, eliminating the requirement of passive LMUs (Wang et al., 2014, 2018).

The Smart-Leg technology is an active method and was firstly applied in 1997 (Labbe and Legras, 1998). The principal idea is to engage the topsides with the substructure at the moment when the heave motion of the vessel reaches the peak and the vertical motion and the kinematic energy are zero. The method is designed to avoid the impact during the mating process, therefore called as shockless floatation method. To achieve the goal of shockless mating, special equipment is designed and installed on both the topsides and on the vessel, including Smart legs, Smart fins, Smart fenders, and Smart shoes. Smart legs installed on the topsides are the most important equipment for mating operations. After the topsides are aligned with the jacket legs, the Smart legs start to work with the jacks to be activated to deploy extension pipes and to mat with the top of jacket legs. Because the surge, sway, and yaw motions are constrained by the Smart fins and Smart fenders, the vessel is free only in the heave direction. A non-return check-valve located on the Smart Leg makes the jack extend or lock into place as the vessel falls with the swell. Once the vessel heave motion reaches the peak, the check-valve closes and locks the topsides with the Smart legs. The velocity of the vessel and the kinetic energy are zero at the moment and the locking action takes place very fast in only a few seconds, no impact loads thus being induced. Then, the topsides and vessel are totally immobilized since the jacket and the vessel now share the topsides load. The load transfer process is finally accomplished by ballasting the barge and collapsing the Smart-Shoes. The technology focuses on handling harsh weather with long period swell and aims at the induced large vertical motions of the floatover installation vessel. However, the technology poses some difficulties on the final elevation and levelness of the topsides and does not allow a single failure. The complex active mechanism and associated specially design Smart components may constrain its applications.



a) DP barge HYSY278



b) Configurations of thrusters and propellers

Fig. 7. DP floatover barge.

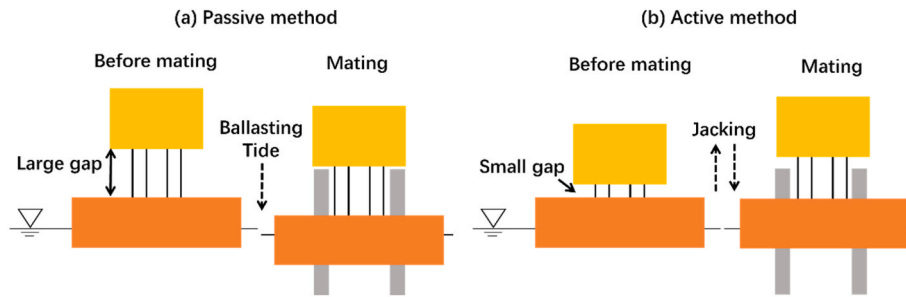


Fig. 8. Illustration of the passive method and the active method.

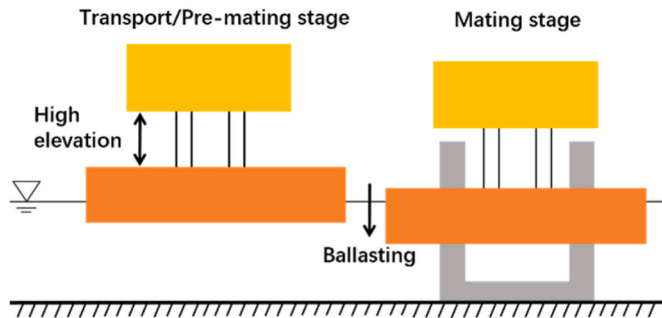


Fig. 9. Illustration of the HIDECK technology.

2.1.3. Classification based on substructure

The floatover installation can be for both the fixed substructure and the floating substructure, as shown in Fig. 12. The fixed substructure includes the GBS and the jacket. The floating substructure includes the spar, semi-submersible, and TLP substructures. Due to the different configuration characteristics of the fixed and floating substructure, the design and requirements of the floatover system are quite different.

The initial floatover method is proposed for the fixed substructure and the first floatover method application is the Maureen steel GBS. The huge GBS can support the mega topsides and are successfully applied in the past decades (Liu and Li, 2017). The topsides are transported over

and matted the shafts of the platform base. Challenges for the GBS floatover operation are obvious. Not only the huge topsides and the GBS are quite heavier, but also the ballasting/de-ballasting performance control is quite complicated. From the view of the hydrodynamic response, the GBS, comprised of several large shafts/columns, has great effects on the hydrodynamic performance of the transportation barges and poses difficulties on the analysis. Sun et al. (2012) and Zhu et al. (2021) have demonstrated the interaction effects due to diffractions by a large volume substructure and the installation barge. The jacket substructures have been developed and widely utilized in the oil and gas industries due to their cost-efficient and convenient installation. The steel-made jacket legs with less volume have enough structural strength for supporting the large topsides and bearing the environmental loads. The jacket configuration can be adopted the slot for the float through of the transportation barge. The fixed substructures also benefit the lash line layout, directly attaching the lash line to the substructure to reduce the motions of the barge and topsides.

The floating platform becomes much more competitive than the fixed platform when it comes to deep water. The fixed substructures are no longer economical and suitable for the deep-sea platform. Floatover installations for the mega topside in the deep water are much more dangerous because of the harsh environment. The floatover installation of the 24000-ton Auger TLP Deck in 1993 is the first open-sea installation onto a floating substructure in place in the Gulf of Mexico (Kenney et al., 1994). The topsides were matted with the columns only 7 m above the water surface of TLP substructures at the mating site with the depth

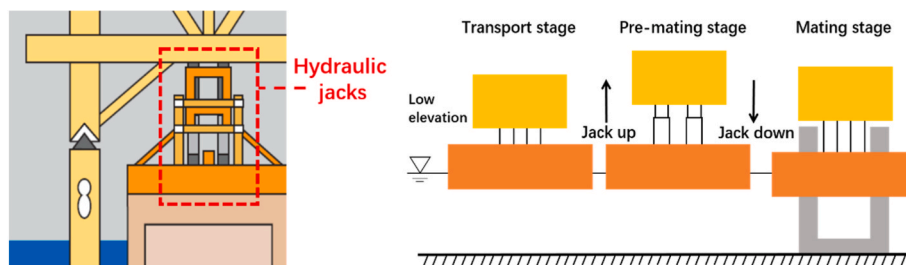


Fig. 10. UNIDECK technology.

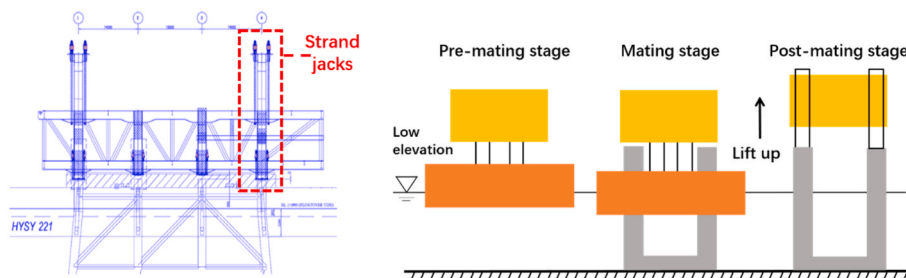


Fig. 11. Strand jack lifting technology.

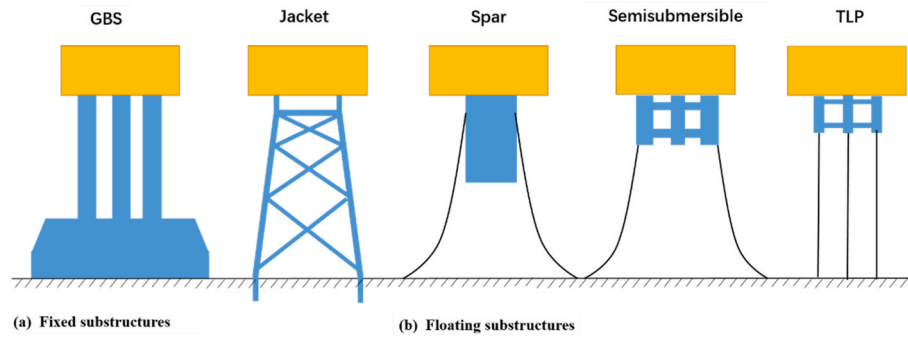


Fig. 12. Schematic of fixed substructures (GBS, jacket) and floating substructures (spar, semisubmersible, and TLP).

of 73 m during the hurricane season, then towing the integrated TLP to the station area. During the mating, de-ballast operations of the TLP hull make the columns rise and attach the topsides legs, therefore few air gaps being needed before the mating. The water plane of the floating substructure acts as a soft spring to absorb the impact load. The requirement for elastomer elements, namely DSU and LMU, can be much more reduced compared to fixed jacket mating installation. For the spar platform, the 3400-ton Kikeh spar topsides floatover installation is the first successful application in which the twin-barge floatover method was employed for the installation as the spar is lack of slots (Edelson et al., 2008). Conducting a floatover operation to the floating hull would face different technical challenges compared to the fixed jacket deck mating installation. The main issue is the motion synchronization of the transportation barge and the floating substructures (Edelson et al., 2007). The relative motions of the mating points in surge and heave motions are strongly dependent on the relative longitudinal and vertical positions of the multiple floating bodies. For reducing the relative motions in a limited range, optimization of the fender, lash and mooring systems should be carried out. The pre-moored substructures with a large mass have less lateral stiffness by the mooring system. The limited stiffness reduces the lateral docking loads and the large mass could generate large impact loads. Numerical analysis of multiple large floating structures is also a great challenge because of the complex hydrodynamic interactions between the transportation barge and the floating substructure, especially for the twin-barge floatover system (Cao et al., 2019; Dessi et al., 2019; Vernes et al., 2019). Model tests are inevitable to accurately estimate the dynamic responses and design loads, thus optimizing the moorings and fenders. Thus, scientific and engineering works on floatover operations are still required for the floating substructures in hash deep water.

2.2. Mating equipment

The LMU and the DSU are the main specialized components for the floatover installation. Fig. 13 illustrates the locations of DSUs and LMUs. They are designed to guide the initial contact and to mitigate the impact loads on barges and topsides. DSUs are designed to support the topsides

during the transportation stage and to absorb the impact loads between the DSF and the mating points during the post-mating stage. Fig. 14 gives the details of the DSU. The DSU consists of the mating pin and the mating receptacle. As shown in Fig. 14, before the separation, the mating pin is contacted with the mating receptacle. After the separation, the mating pin is separated from the mating receptacle while the prepared sands in the mating pin drop into the cone of the mating receptacle. The sands can alleviate the steel-to-steel impact loads on mating points.

LMUs are designed to buffer the impact load between the support receptacles and the stabbing pins during the mating stage. LMUs are illustrated as shown in Fig. 15(a) and Fig. 15(b). LMUs help make the initial contact smooth and reduce relative motions of the stabbing pins and receptacles before engagements. Fig. 15(c) illustrates the components of the LMUs. As the shock absorber, the LMU is composed of the rubber and the sand tank under the receptacles. The units are custom designed to balance leg reaction load. The stiffness of the LMU is linear to great extent, as shown in Fig. 3(d). As different materials and configurations of the LMU, the vertical and lateral strength should be examined by experimental and numerical simulations before the practical applications.

Loads on mating units are crucial for the design of the mating units. Research on the design loads on mating units was published recently. Kagita et al. (2019) numerically simulated the rapid jack-down operations and evaluated the impact loads between the jacket and topsides deck. Kim et al. (2019a) and Choi et al. (2014) carried out the numerical modeling and performed a series of parametric studies to investigate the effect of reaction force and maximum stress occurring of LMU during the mating procedure of the topsides on a semi-submersible substructure. Hu et al. (2016) numerically investigated the floatover installation of the semisubmersible platform topsides and evaluated the pressure of the loads on LMUs. The nonlinear impact loads on the LMUs and the effect of LMU angle on the impact loads were discussed in terms of the bifurcation phenomena and maximum impact loads on LMUs. Bai et al. (2021a) experimentally examined the loads on LMUs and DSUs during the twin-barge floatover installation, as shown in Fig. 16. Loads on mating units in the rapid floatover installation are of practical interest and are

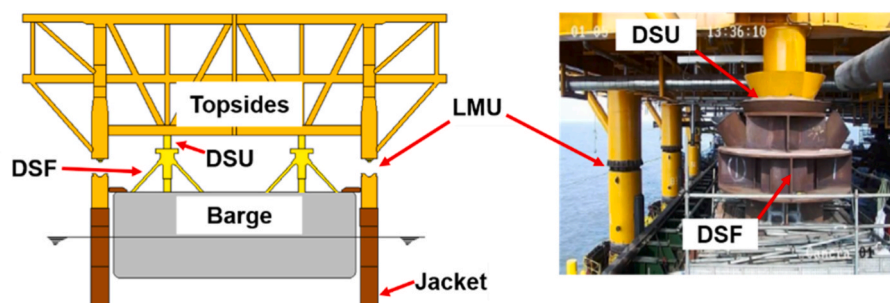


Fig. 13. Configurations of DSUs and LMUs.

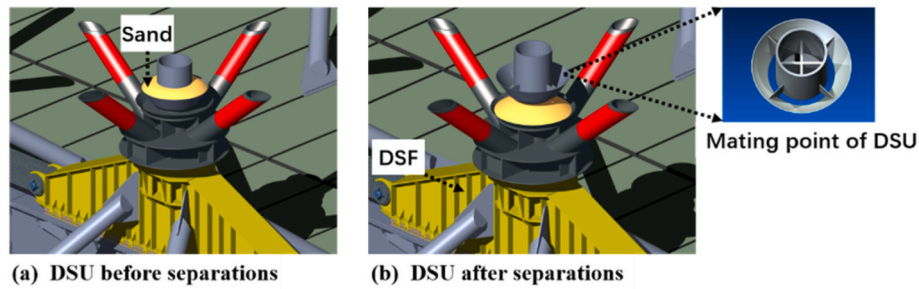


Fig. 14. DSU.

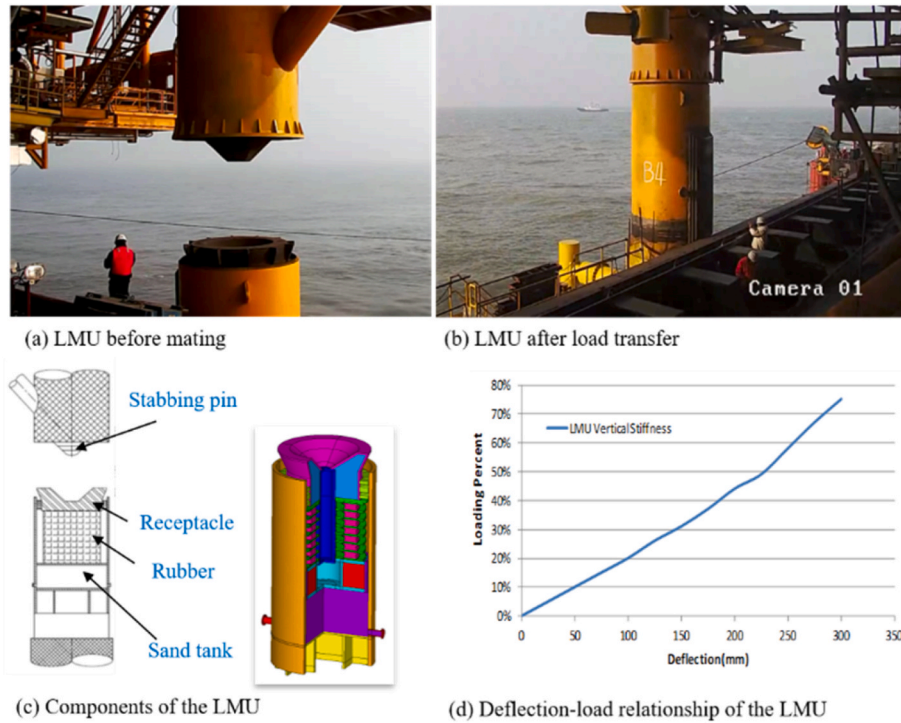


Fig. 15. LMU.

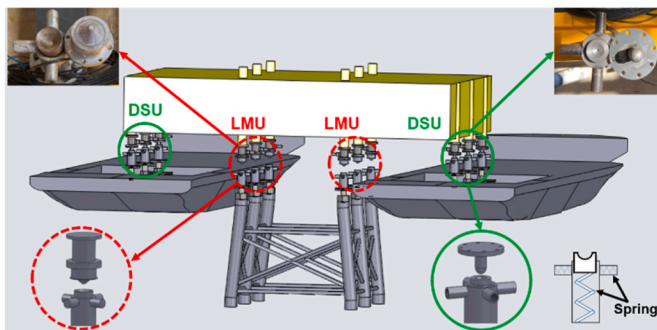


Fig. 16. Experimental model of LMU and DSU.

associated to the design of the jacking system. Bai et al. (2020b) experimentally simulated the floatover installation system using the rapid load transfer technology and examined the vertical and lateral loads on LMU and the jacking system. The experimental results indicated the jacking system shall provide lateral restraint equivalent not less to 20% of the structure weight acting horizontally, rather than the suggested 10% of the structure weight by GL Noble Denton Guideline

(2015). Wang et al. (2018) presented the strand-jack floatover installations and discussed the capability of the rapid floatover method using the strand jacks. Cho et al. (2020) outlined the feasibility of an active jacking system for floatover operation at the swell wave area based on the model test result to compensate the vertical vessel motions during docking operation stages.

Different configurations of mating units are customized for different applications. The cost and the safety of the mating units are crucial for the floatover operations. The structural design and optimization of the mating units motivate researchers and engineers. Tan et al. (2008) addressed design issues and technical challenges of mating components for the Arthit project during the loadout, transport, and floatover operations, in which the full-scale load testing program of elastomer pads and assemblies was conducted. Chaitanya and Nair (2013) addressed the structural behaviors of the LMU under the combination of vertical and horizontal loads during the mating process based on the nonlinear finite element analysis and provided suggestions on the selection of LMU elastomer. Lee et al. (2018) provided guidelines on the design of LMU composed of hyperelastic materials for the floatover installation of 20000-ton topsides. They applied the Mooney-Rivlin material model to express the behavior of LMU elastomer and performed static finite element analysis with the considerations of large deformations. With aim of minimizing the total mass and the maximum von-Mises stress of

LMU, the design optimization for multi-objective functions based on the design sensitivity analysis were presented. Song et al. (2020) derived the DSF stress distributions based on the three-dimensional finite element analysis. They performed sensitivity analysis using the orthogonal array of DOE on the DSFs. The main structural member dimensions of the DSFs were considered as the three-level design variables, and the maximum stress and weight were set as response functions. The passive type DSF and the active type DSF were examined with the considerations of the loads during the floatover operations for the safe offshore operation.

Those studies indicate that the loads on units varied with the topside weight as well as the load transfer stage, which requires the dedicated experimental and numerical analysis for each floatover installation. Individual elastomer elements and assemblies of mating units must be tested to ensure functionality and structural integrity under full design loads. Factors affecting the fabrication and assembly of the hardware components must be considered in the design and analysis.

2.3. Standards and guidelines

The topsides installation must follow the guidelines of regulatory authorities to ensure safety. The risk management, operation criteria, weather forecast, and structural integrity during the floatover installations are required by the offshore oil and gas industry. DNV (2014) and ISO 19901-6 (2009) provide general guidance on marine operations of various types of offshore structures, including subsea templates and foundations. The simplified formulas for establishing design loads for planning and execution of marine operations are suggested. Those guidelines cover load-out operations, towing operations, and weather criteria. The DNV (2012) standard provides requirements for offshore load transfer operations, including the ballasting operations and the mating operations. GL Noble Denton Guideline 0031/ND (2015) provides guidelines for the floatover installation on both fixed substructures and floating structures. DNVGL (2016a) and DNVGL (2016b) provide comprehensive information on various offshore operations and address the site-specific assessment for different operations. CCS (2020) guidelines provide detailed requirements on the transportations and floatover operations of the fixed platform. Although various standards and guidelines are provided, improvements on floatover operations are always needed. Further, as novel floatover techniques applied for the marine operation activities are not covered by standards and guidelines either, industry knowledge, experience, and technical know-how should be exercised for evaluating new installation concepts.

3. In-site measurement

The in-site monitoring technologies are indispensable and have substantially reduced the risk of practical floatover operations. The in-site monitoring system has been proven an effective method for obtaining real-time tracking and feedback information, providing necessary operational support, minimizing any potential operation risks, and enabling offshore operations as smoothly and efficiently as possible. Benefited from the development of the field monitoring technologies for offshore platforms (Peters et al., 1990; Edwards et al., 2005; Goldsmith et al., 2007; Peng and Zhi, 2012; Russell, 2017), the in-site monitoring for the floatover installations have been well established. Researchers (Luo et al., 2015; Yu et al., 2017; Zhu et al., 2017; Wang et al., 2018) presented the successful application of the monitoring system in the floatover installation, in which the real-time information was obtained instantaneously to guide the floatover operations. Fig. 17 illustrates the monitor system. The utilized comprehensive field measurement program consisted of several sub-system parts. 1) The motion monitoring system includes an integrated INS and GPS, used as mobile units, as well as a single RTK base station installed on a fixed platform nearby, thus providing up to centimeter-level accuracy to monitor the six-degree-of-freedom barge motions and the movements of topsides LMU cones and tips of DSU. 2) The environmental monitoring systems include

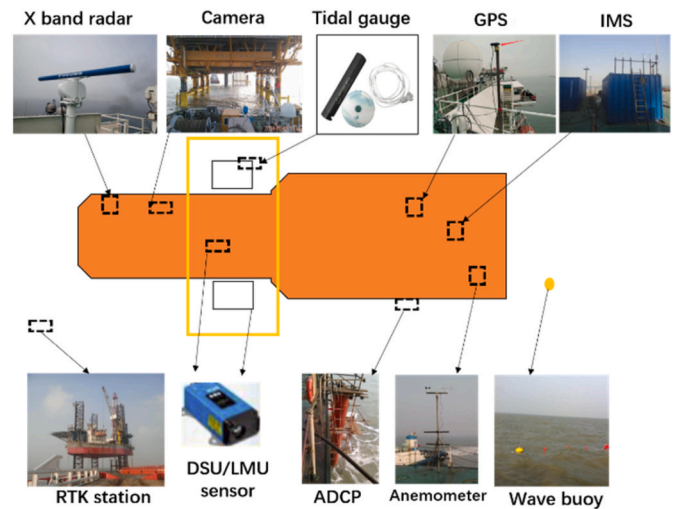


Fig. 17. Monitoring system.

one buoy system for the wave height, period, and direction, the ADCP for the profile current speed and direction, the anemometer, and the tidal gauge. The wave measurement was further improved by X-Band Wave Radar together with Vertical Measuring Range Finder (Wang et al., 2017; Tian et al., 2018). For the tide monitoring, it is usually started four and more days prior floatover installation. Comparison between the tide measured data and the weather forecast is carried out and helps to calibrate the safe clearance between the topsides LMUs and the jacket legs. Finally, the draft of the DTV is adjusted before docking to compensate for this discrepancy. For wave measurements, the significant wave heights are calculated based on the most recent dozen minutes. Once the maximum significant wave height is lower than the required value, the floatover installation is able to be conducted. The wind speed is also monitored before floatover installation to help make the decisions of the floatover installation. 3) Mating and separating monitoring system. The visual monitoring system with high-resolution video cameras is installed on the deck support frames to observe the mating operation of the LMU and the separation of the DSU. The stress and loads monitoring were also carried out to measure the stress of the LMU and the jack during the jacking operations (Yang et al., 2014). Tian et al. (2018) addressed the integrated monitoring system and presented the successful applications in Malaysian, in which multichannel monitoring data was collected with a variety of remote signal transmission technologies and data sharing technology. The integrated monitored information could be transported from the service center to the bridge for operators with the wireless method. The movement measure of the LMU and DSU points are of great importance for the decisions of the ballasting/jacking operations. Several video cameras for all mating points are applied for providing real-time visual information during the mating operations. The motions of the mating points less than the capture radius of the mating units can make the mating operations smooth.

From the authors' point of view based on our practical experience, the main challenge of the in-site measurement is the reliability and stability of the monitoring system as the in-site measurement system is quite complicated with lots of distributed sensors, a large amount of measurement information, and the large number of the collaborative engineers. The design of the monitoring system should be optimized to make it more intelligent and systematic. The measurement accuracy, data synchronizations, and data integrate should be improved to reduce the human interventions and the number of field engineers.

4. Experimental modeling

Because of complex configurations and dynamic characteristics of

the floatover installation, physical model tests are required, especially for the novel floatover method and the mega topside installation. Most analysis focus on the mating stage as the mating phase is most critical for the floatover operations. The mating phase is generally considered as several stationary stages. The entire load transfer process, from the initial contact stage (0% stage) to the final separate stage (100% stage), is assumed as several steady-state intermediated stages. During the load transfer process, the topside and the barge compose a rigid body before the mating operations and then consist of the flexible connected body at the middle stages. Finally, the topside and the barge are two separated bodies when the load transfer process is accomplished. In consequence, no less than three representative scenarios should be considered to evaluate the feasibility of the entire load transfer process.

Researchers and engineers have conducted a lot of experiments based on the steady-state assumption to explore the feasibility of different floatover systems. Koo et al. (2010a) carried out a series of experimental investigations including 0%, 20%, 80%, 100% stages on the twin-barge floatover system for the spar topsides, in which the stationary motions of the barges and the topsides at each stage were analyzed. The relative motions between the topsides and the spar were statistically evaluated as well as the fender and mating loads. Kim et al. (2017, 2019) and Kwon et al. (2019) considered three mating stages of 10%, 40%, 70% load transfers for the fixed jackets and two mating stages of 10%, 80% load transfers for the semi-submersible and discussed the effects of wave headings, load transfers and topside center of gravity on the mating units impact forces. Li et al. (2020) and Bai et al. (2020a) investigated the 0%, 30%, 70%, 100% stages of the load transfer process of a T-shaped floatover system, in which the variations of the topside-barge posture were detailed. Bai et al. (2021a) developed the elastically-connected twin-barge floatover model and investigated the steady-state motions behaviors at the 0%, 50%, and 100% stages in various wave headings. The hydrodynamic shielding effect in beam and quarter seas of the upwind barge on the leeward barge and the potential gap resonance of the in-between fluids were roughly estimated based on the formula proposed by Lewandowski (2008).

As the load transfer operations are continuous with variable barge drifts, the un-steady continuous experimental modeling is more accurate and close to realistic situations. Magee et al. (2014) simulated the ballasting barge by changing the elevation of the barge relative to the jacket by reducing the water depth in the experimental tank, demonstrating the continuous barge motions and mating loads. However, the change in the barge mass and the effect of water depth on the wave elevations were not considered. Koo et al. (2010a) experimentally simulated the continuous load transfer process from 20% to 50% of the twin-barge floatover installation of spar topsides by vertically jacking down the topsides, in which the time-varying statistics of the relative motions and the decreased standard deviation of the motions response were found. Bai et al. (2020a) developed the controllable continuous floatover installation model of the rapid jacking floatover system with the T-barge, in which the jacking system with the controllable jack stroke and the ballasting system with the variable ballasting pump were employed to accurately adjust the posture of the floatover system. The continuous load transfer process from the 0% stage to the 100% stage was simulated repeatedly with different random wave seeds to achieve the load and motion characteristics of the rapid floatover system. Comparing the results of mating loads with those from the steady-state experimental modeling indicated that the steady-state modeling may underestimate the mating loads during the rapid and continuous load transfer process. Tao et al. (2021) used the controllable pumps injecting water into the distributed ballast tanks on the barges to change the draft of barges automatically and continuously for the twin-barge floatover system. The dynamic motion response of the topside, transportation barge, and installation barges were found to be the multi-modal non-Gaussian response, and there exist significant yaw-surge/yaw-sway coupling effects between the transportation barge and installation barges.

Apart from the steady-state assumptions, there are various simplifications to a certain extent in the above experiments. For example, the practical rubber fender with the nonlinear load-deflection characteristic is generally assumed as a linear spring, which may result in unrealistic fender forces. The experimental modeling of the mating units also should be devoted to efforts because in most the experiments neither the nonlinear load-deformation relationship nor the frictions between the mating pins and the mating cone are modeled. For the freely floating twin barges and the substructures in beam and quarter seas, there are multiple gaps between the barges and between the barge and substructures, potentially leading to the gap resonances. However, the experimental measurements and assessments of the gap fluid effect are difficult due to the complex floatover system configurations and the drift motions. Experimental identifications of the multi-body hydrodynamic interactions as well as the mechanical interactions are challenging and required to be further experimental investigations.

5. Numerical modeling

Numerical modeling is indispensable for the design of the floatover system. Various numerical tools have been developed to investigate the dynamic response of the floatover installation within the framework of potential flow theory, including the commercial software and the in-house codes.

Similar to the experimental modeling, several steady-state intermediated stages of the floatover installation are assumed. Tahar et al. (2006) numerically analyzed 15% and 85% stages with the aims of verifying the synchronizations and desynchronizations of the topside and the barge for the rapid ballasting operations. Jung et al. (2018) carried out the numerical analysis of the mating stage at 0%, 50%, and 100% stages. Koo et al. (2010b) carried out the numerical analysis by the proprietary software FREDOM and MULTISIM for the twin-barge floatover installation of spar topsides at 0%, 20%, 30%, and 80% stages to identify hydrodynamic and mechanical interactions between twin-barges and spar. In those simulations, the hydrodynamic problem is solved first, obtaining the wave excitation force and the added-mass and damping coefficients in the frequency domain at each stage. The time-domain motion equations are then formulated by the coevolution approach described by Cummins (1962). The connections between the topside and the barge as well as the external components (mooring and fenders) are simulated.

However, the load transfer process is continuous and even rapid. The time-varying draft and mass of the barge make the stationary assumptions challenging. Kagita et al. (2019) focused on the rapid jack-down operations by the hydraulic jacks and numerically investigated the loads on mating units during the rapid jack-down based on the software MOSES. However, little information on the time-varying positions of the topsides and barge are included. The time-varying characteristics of the multi-body can result in quite different motion and load behaviors of the floatover system as demonstrated in the physical model tests in Bai et al. (2020a). Tao et al. (2020) and Zhao et al. (2021) proposed a simulation method of the continuous load transfer process of the twin-barge floatover installation by the rapid ballasting system, in which the fast interpolation using the established database of barge-draft dependent hydrodynamic coefficients was adopted to evaluate the time-varying hydrodynamic forces. The real-time radiation hydrodynamic coefficients are linearly interpolated at the new average hull wetted surface, while the first-order wave loads are linearly interpolated at the instantaneous hull position. However, the experimental validations of the continuous numerical model and the calculations of the associated retardation function are still insufficient. The continuous load transfer process also involves the transformation of the topside-barge-substructure from the separated multi-body to the elastically-connected multi-body until the rigid-connected multi-body. Those issues pose great difficulties for accurate estimations of the dynamic response of the continuous load transfer process.

Another challenge for the numerical modeling of the floatover system is the hydrodynamic and mechanical interactions of the multiple fixed and floating bodies with considerations of the lash, fender, and mating systems. There have been many attempts to model complex multi-body systems with different simplifications. Sun et al. (2012) numerically investigated the hydrodynamics of the multi-body system comprised by the installation barge with the fixed substructures evaluated here by the computer program DIFFRACT. In the simulation, three different configurations of the substructures were assumed as idealized horizontal constraints: the slender substructures, the columns, and the columns plus caissons although the lash system was simplified as idealized rigid constraints. The former was without considerations of the effect of substructures on the wave field and the latter two were with the scattered and radiated wavefield from the barge being diffracted by substructures. They found that the horizontal constraints of the substructures introduced larger roll motions of the installation barge and showed the large magnifications in the local wave elevation at the midship position on the upwind side of the barge in beam seas. Ma et al. (2018) numerically analyzed the fender collision forces during the floatover installation, in which the hydrodynamic model was established based on the software SESAM with considerations of the second-order waver drift forces. Hu et al. (2017) evaluated the nonlinear impact loads on the mating units at the initial contact stage, in which the state-space model was developed to replace the time-consuming convolution in calculating the retardation functions. The 3-degree sway-roll-heave model of the floatover system was developed to investigate the impact loads at the 0% stage and the sub-harmonic, quasi-periodic, and chaotic responses were numerically investigated. Chen et al. (2014, 2017, 2019) developed a coupled heave-roll-pitch model of the floatover system with the state-space approach to evaluate the vertical impact loads on LMU and DSU at both the 0% and 100% stages, in which the connections between the fixed substructure, the barge, and the topsides were modeled as linear spring-damper systems. Jung et al. (2009) modeled the LMU based on the software MULTISIM (Paulling JR, 1995) as the bi-linear and nonlinear spring, considering the geometrical information and the varied stiffness model of the mating cone, in which the fender and LMU vertical forces were evaluated. Kwak et al. (2018) assessed the effects of the gap between DTV and fender, fender stiffness, and mooring line stiffness on the dynamic response of the floatover system based on the software AQWA. Cao et al. (2019) numerically simulated the twin-barge and the spar topsides at the 50% stage with rigid and hinge connections based on the software AQWA and demonstrated that rigid connections greatly enlarged the mating forces compared with the hinge connections. Xu et al. (2015) investigated the hydrodynamic responses of the three side-by-side barge in the floatover operations in frequency and time domain analysis at the 0% stage by the in-house codes, in which the overestimated hydrodynamic coefficients in frequency domain calculated from the potential flow theory caused by the gaps between the barges were improved by the damping lid method. The unrealistically high peak values of the hydrodynamic coefficients near the resonance frequencies were suppressed successfully by the damping lid method, thus avoiding the exaggerated motions of the three barges in the time domain.

As can be seen from the literature review given above, the commercial software AQWA, SESAM, and MOSES are widely employed for the conventional steady modeling method. However, the simulations based on the commercial software are simple and idealized, in which some aspects are ignored, such as the nonlinearity of the mating unit and fender, the nonlinearity of the wave loads, and the gap fluids resonance. The accurate estimations of the dynamic response of the floatover installation by the commercial software are still unavailable because of lots of the simplifications although the commercial software can provide an approximate result. Researchers also try to develop the advanced proprietary software and code to simulate the floatover installation. Different codes focus on different issues. Some aspects, such as the hydrodynamic radiation and diffraction problem of the multi-body system,

the vertical impact on mating units, the connection stiffness of the multi-body, are investigated in detail with more advanced algorithms and more accurate models. Unfortunately, the comprehensive floatover model with more details and high accuracy is still challenging. Moreover, the validated continuous load transfer process model of the floatover installation is rarely reported, neither by the commercial software nor by the proprietary code. Therefore, more research and development studies should be carried out on the numerical model of the floatover system.

6. Conclusions remarks

This paper provides a comprehensive review of the floatover installation technologies for the platform topsides. The study firstly presents the classifications of various existing floatover methods based on the barge, the transfer system, and the substructure. The developments and the academic research for each method are critically reviewed, and merits along with challenges of each method are discussed. The specialized mating equipment for the floatover installation and the associated standards are surveyed. Finally, investigations by the in-site measurement, the experimental modeling, and the numerical modeling are presented. The challenges and shortages within the emerging research field are highlighted. The main conclusions are as follows.

- The floatover installation methods for the platform topsides have reached a degree of maturity and are successfully applied in practical engineering projects. However, the increasing topsides weights and the complicated environments in harsher offshore areas pose great difficulties on the existing floatover installation methods. Although a few novel concepts have also been suggested for offshore topsides installations, most concepts still have low technology readiness levels. Novel and reliable floatover methods are still open, especially for the mega topsides of the floating platform. The following technical method related to the topside floatover installation are envisaged to experience rapid advancement in the future: (a) floatover method with the specially designed barge. The T-shaped/twin barge accommodated to the substructures lack of slots and mega topsides with large weights may make the complex topsides installation available and release the limitation of the substructures and the topsides. (b) rapid floatover method with the active jacking system. There is a growing need for a reliable and efficient jacking system with aim of reducing the mating time and alleviating the impact loads. The active and controllable jacking system makes floatover operations less time-expensive compared with the conventional passive floatover operations, thus reducing operational costs. (c) floatover method assisted by the dynamically positioning system and other motion compensation systems. Those systems are expected to evolve for limiting barge and topsides motions, benefiting the most dangerous mating operations.
- Specialized mating units for various floatover methods are critical for floatover installations. However, no consensus has been reached on the design of the mating units which are generally assembled by steel and composite materials and with highly nonlinear characteristics. The existing standards provide less information on mating units. Successful standardization of the industrial products can lead to novel floatover operations under more weather windows, less human intervention, and lowered risks for offshore installation. In addition, the robust experimental and numerical models of the mating units are still limited, considering the conventional linear assumptions in the analysis. There is still a need to develop models for testing, designing, and simulating the units.
- General purpose codes require substantial extensions to capture the continuous unsteady-state floatover process although the conventional hydrodynamic programs are applied for simulating the hypothetical steady-state floatover stages. The interactions of the

multibody, including the topsides, mating units, barges/catamaran, mooring lines, lashing lines, and fenders, need to be reasonably analyzed with the considerations of the nonlinear behavior of the components. A desirable numerical tool for such purposes should be modularized and preferably open-source.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Abbot, P.A., Farmer, L.E., Blight, G.J., Osborne-Moss, D., Wimpey, G., 1980. A New Integrated Deck Concept. Offshore Technology Conference, Houston Texas.
- Bai, X.D., Luo, H.B., Xie, P., 2020a. Effect of wave headings on the dynamic response of the continuous mating operation of floatover installation. *China Ocean Eng.* 35 (1), 72–83.
- Bai, X.D., Luo, H.B., Xie, P., 2020b. Experimental analysis of dynamic response of floatover installation using rapid transfer technique in continuous load transfer process. *J. Mar. Sci. Technol.* 25, 1182–1198.
- Bai, X.D., Luo, H.B., Xie, P., 2020c. Experimental investigation on motions behavior and loads characteristic of floatover installation with T-shaped barge. *Ocean. Eng.* 195, 116761.
- Bai, X.D., Luo, H.B., Xie, P., 2021a. Experimental analysis of hydrodynamic response of the twin-barge floatover installation for mega topsides. *Ocean. Eng.* 219, 108302.
- Bai, X.D., Luo, H.B., Xie, P., 2021b. Experimental investigation on hydrodynamic performance of side-by-side three barges in topsides floatover installation in beam seas. *Ocean. Eng.* 236, 109516.
- Bai, X.P., Li, D., Fan, M., Yi, C., 2014. Float-over installation of Huizhou 25-8 DPP topside using dynamically positioned vessel. *Ocean. Eng.* 32 (5), 70–77 (in Chinese).
- Bakar, A.A., 2019. Successful Float over Installation by Portable DP Vessel. Abu Dhabi International Petroleum Exhibition and Conference, Abu Dhabi, UEM.
- Beal, D.D., Allen, J.D., Datta, B.N., 1992. Design and Installation of the North Dauphin Island Platform. Offshore Technology Conference, Houston, Texas, USA.
- Beerendonk, M., Groves, N., Palmer, R., 2008. Float-over operations using dynamically positioned vessels. In: Proceedings of Royal Institution of Naval Architects International Conference. London, UK.
- Bokhorst, J., Willemse, O., Zontjes, R., 2011. Float-over analysis for world's largest float-over barge. In: Proceedings of the 30th International Conference on Offshore Mechanics and Arctic Engineering. Rotterdam, The Netherlands.
- Cao, A.X., Sang, S., Sheng, J., Yu, S.N., Zhang, G.M., Shi, X., 2019. Numerical simulation of twin-barge float-over installation on AQWA software. *Cluster Comput.* 22, 8735–8746.
- Cummins, W., 1962. The impulse response function and ship motions. *Schiffstechnik* 9 (1661), 101–109.
- CCS, 2020. Guidance Notes GD03-2020, Guidelines for Transportations and Floatover Operations of Marine Structures.
- Chaitanya, K., Nair, S.B., 2013. Design of leg mating unit for float-over installation of decks. In: Proceedings of the 32th International Conference on Offshore Mechanics and Arctic Engineering. Nantes, France.
- Chen, M.S., Taylor, E.R., Choo, Y.S., 2014. Time domain modeling of a dynamic of a dynamic impact oscillator under wave excitations. *Ocean. Eng.* 76, 40–51.
- Chen, M.S., Taylor, E.T., Choo, Y.S., 2017. Investigation of the complex dynamics of float-over deck installation based on a coupled heave-roll-pitch impact model. *Ocean. Eng.* 135, 262–275.
- Chen, M.S., Wang, F., Zhu, L., Pedersen, P.T., 2020. Dynamic analysis of offshore wind turbine installation based on a novel integrated mating method. In: Proceedings of the 30th International Offshore and Polar Engineering Conference (Shanghai, China).
- Chen, M.S., Zou, M.Y., Zhu, L., Sun, L., 2019. Numerical analysis of GBS float-over deck installation at docking and undocking stages based on a coupled heave-roll-pitch impact model. In: Proceedings of the 38th International Conference on Offshore Mechanics and Arctic Engineering. Glasgow, UK.
- Cherian, L.D., Suresh, P.K., Takiyeddine, O.H., 2014. Topside Installation by Float over - Engineering Challenges. Abu Dhabi International Petroleum Exhibition and Conference, Abu Dhabi, UAM.
- Cho, C.G., Son, J., Nam, H.W., Sung, H.G., Lee, K.S., 2020. Model test of active jacking system for float-over operation at swell area. In: Proceedings of the 30th International Offshore and Polar Engineering Conference (Shanghai, China).
- Choi, Y.M., Nam, B.W., Hong, S.Y., 2014. Numerical modeling of load transfer unit of the float-over installation. In: Proceedings of the 24th International Ocean and Polar Engineering Conference (Busan, Korea).
- Cholley, J.M., Cahay, M., 2007. Float-over High Air Gap: New Solution. Offshore Technology Conference, Houston, Texas, USA.
- Cholley, J.M., Cahay, M., 2009. Offshore wind turbine and substructure modelling for float over installation design. In: Proceedings of the 28th International Conference on Ocean. Offshore and Arctic Engineering, Honolulu, Hawaii, USA.
- Cholley, J.M., Delage, B., Cahay, M., 2009. Float-over at high air gap design and issues. In: Proceedings of the 36th International Conference on Ocean. Offshore and Arctic Engineering, Honolulu, Hawaii, USA.
- Dessi, D., Faiella, E., Pigna, C., Celli, C., Miliante, T., Di, P.E., 2019. Experimental analysis of the seakeeping response of a double-barge floatover system for topside transportation. *Int. J. Offshore Polar Eng.* 29 (2), 198–206.
- Dessi, D., Faiella, E., Pigna, C., Celli, C., Miliante, T., Di, P.E., 2017a. Experimental analysis of topside transportation with a double-barge float-over system. In: Proceedings of the 27th International Ocean and Polar Engineering Conference. Sapporo, Japan.
- Dessi, D., Faiella, E., Pigna, C., Celli, C., Miliante, T., Di, P.E., 2017b. Experimental analysis of the station keeping response of a double-barge float-over system with an elastically scaled physical model. In: Proceedings of the 27th International Ocean and Polar Engineering Conference. Sapporo, Japan.
- DNVGL, 2016a. DNVGL-ST-N001, Planning and Execution of Marine Operations.
- DNVGL, 2016b. DNVGL-ST-N002, Site Specific Assessment of Mobile Offshore Units for Marine Warranty.
- DNV, 2014. DNV-OS-H101, Mairine Operations.
- DNV, 2012. DNV-OS-H201, Load Transfer Operations.
- Duquesnay, J.L., Baldwin, J., Rains, J.W., 2013. Docking and undocking considerations for floatover analyses and operations. In: Proceedings of the 36th International Conference on Ocean, Offshore and Arctic Engineering. Nantes, France.
- Edelson, D., Luo, M.Y., Halkyard, J., Smiley, D., McFadyen, M.K., 2008. Kikeh Development: Spar Topside Floatover Installation-Naval, Offshore Technology Conference. Offshore Technology Conference, Houston, Texas, USA.
- Edelson, D., Halkyard, J., Chen, L.Y., Chabot, L., 2007. Floatover deck installation on spars. In: Proceedings of the 26th International Conference on Offshore Mechanics and Arctic Engineering. California, USA. San Diego.
- Edwards, R., Prislín, I., Johnson, T., Campman, C., Leverette, S., Halkyard, J., 2005. Review of 17 real-time, environment, response, and integrity monitoring systems on floating production platforms in the deep waters of the Gulf of Mexico. In: Proceedings of the 20th International Conference on Information Systems.
- Geba, K., Welaya, Y., Lehet, H., 2017. The hydrodynamic performance of a novel float-over installation. *Ocean. Eng.* 133, 116–132.
- GL Noble Denton, 0031/ND, 2015. Guidelines for Float-Over Installations.
- Gros, G., Lescurat, A., 1982. Deck mating-installation of heavy packages on offshore structures. In: Offshore South East Asia Show. Singapore. SPE-10475-MS.
- Goldsmith, B., Foyt, E., Hariharan, M., 2007. The role of offshore monitoring in an effective deepwater riser integrity management program. In: Proceedings of the 26th International Conference on Offshore Mechanics and Arctic Engineering. California, USA. San Diego.
- He, H.C., Wang, L., Zhu, Y.M., Xu, S.W., 2020. Numerical and experimental study on the docking of a dynamically positioned barge in float-over installation. *Ships Offshore Struct.* <https://doi.org/10.1080/17445302.2020.1737452>.
- Hu, Z.H., Li, X., Li, J., Wu, X., Zhang, F., 2016. Nonlinear dynamics and bifurcation phenomena in float-over systems under wave excitations. In: Proceedings of the 26th International Ocean and Polar Engineering Conference (Rhodes, Greece).
- Hu, Z.H., Li, X., Zhao, W.H., Wu, X., 2017. Nonlinear dynamics and impact load in float-over installation. *Appl. Ocean Res.* 65, 60–78.
- ISO 19901-6, 2009. Petroleum and natural gas industries - specific requirements for offshore structures - Part 6. Mar. Oper.
- Jin, X.J., Wang, A.M., Li, H.L., Yu, W.T., He, M., Wang, A., 2018. Floatover installation technology with a DP2 class dynamic-positioning semisubmersible vessel. In: Proceedings of the 28th International Offshore and Polar Engineering Conference, Sapporo, Japan.
- Joshi, R.P., 2018. Offshore and Marine: Challenging Offshore Installation Tactics and First Float over Installation Accomplishment in Cheleken Field. Abu Dhabi International Petroleum Exhibition and Conference, Abu Dhabi, UEM.
- Jung, J.J., Lee, W.S., Shin, H.S., Kim, Y.H., 2009. Evaluating the impact load on the offshore platform during float-over topside installation. In: Proceedings of the 19th International Offshore and Polar Engineering Conference. Osaka, Japan.
- Jung, S.J., Kwak, H.U., Oh, S.H., Kwon, Y.J., Nam, B.W., Kim, N.W., Lee, K.S., Sung, H. G., 2018. Mating analysis for float-over installation of a large topside in various load transfer stages. In: Proceedings of the 28th International Offshore and Polar Engineering Conference. Sapporo, Japan.
- Kagita, G., Addal, M.B., Achary, G.G.S., Sripada, S.A.R., 2019. Evaluation of impact loads on offshore jacket platform during float-over mating operation. In: Proceedings of the 38th International Conference on Offshore Mechanics and Arctic Engineering. Glasgow, UK.
- Kenney, J.J., Knoll, D.A., Brooks, A.I., 1994. Loadout of Auger Deck and Hull-Deck Mating Operations. Offshore Technology Conference, Houston, Texas, USA.
- Kim, H.S., Lee, K.S., Park, B., Sung, H.G., 2019a. Design of elastomeric pad in LMU for semi-submersible float-over installation. In: Proceedings of the 29th International Ocean and Polar Engineering Conference (Honolulu, Hawaii, USA).
- Kim, N.W., Nam, B.W., Kwon, Y.J., Cho, S.K., Park, I.B., Sung, H.G., 2017. Experimental investigation on mating operation during float-over installation of a large topside in waves. In: Proceedings of the 26th International Ocean and Polar Engineering Conference (San Francisco, California, USA).
- Kim, N.W., Nam, B.W., Kwon, Y.J., Park, I.B., Cho, S.K., Sung, H.G., 2019. An experimental study on the float-over installation of a semi-submersible. In: Proceedings of the 29th International Ocean and Polar Engineering Conference (Honolulu, Hawaii, USA).
- Kocaman, A., Kim, D.J., Seto, J., 2008. Float-over of Arthit PP Deck. Offshore Technology Conference, Houston, Texas, USA.
- Koo, B., Magee, A., Lambrakos, K., Beyko, E., Sablok, A., 2010a. Prediction of motions and loads for floatover installation of spar topsides. In: Proceedings of the 29th International Conference on Offshore Mechanics and Arctic Engineering (Shanghai, China).

- Koo, B., Magee, A., Lambros, K., Beyko, E., Sablok, A., 2010b. Model tests for floatover installation of spar topsides. In: Proceedings of the 29th International Conference on Ocean, Offshore and Arctic Engineering (Shanghai, China).
- Kwak, H.U., Jung, S.J., Oh, S.H., Kwon, Y.J., Nam, N.W., Kim, N.W., Lee, K.S., Sung, H. G., 2018. Assessment of docking operation during float-over installation on jacket structure. In: Proceeding of the 27th International Ocean and Polar Engineering Conference. Sapporo, Japan.
- Kwon, Y.J., Nam, B.W., Kim, N.W., Park, I.B., Sung, H.G., 2019. Experimental and numerical study on docking operations of a transportation vessel for float-over installation. *Int. J. Offshore Polar Eng.* 29 (2), 207–216.
- Labbe, J.P., Legras, J.L., 1998. Smart-Leg: A New Method for Heavy Deck Offshore Installation. Offshore Technology Conference, Houston, Texas, USA.
- Lee, K.S., Park, B., Kim, H.S., Kim, D., Sung, H.G., 2018. Design of leg mating unit for float-over installation. In: Proceedings of the 27th International Ocean and Polar Engineering Conference. Sapporo, Japan.
- Lewandowski, E.M., 2008. Multi-vessel seakeeping computations with linear potential theory. *Ocean. Eng.* 35, 1121–1131.
- Li, D., Liu, T.J., Ji, H., Ran, A., Kim, M.H., Kang, H.Y., 2015. Development and application of a DP float-over analysis program. In: Proceedings of the 25th International Ocean and Polar Engineering Conference. Hawaii, USA.
- Li, H.L., Luo, H.B., Bai, X.D., Xie, P., Qin, L.C., Zhu, S.H., 2020. Experimental investigation on floatover installation using rapid load transfer technique for one mega topside in beam waves. In: Proceedings of the 30th International Ocean and Polar Engineering Conference. Shanghai, China.
- Lin, D., Zhang, D., Cho, C.G., Wang, A., Ruskin, A., Qiu, X.H., 2016. Overcoming challenges to deliver West Africa's first DP floatover topsides installation. In: Proceedings of the 26th International Ocean and Polar Engineering Conference. California, USA. San Francisco.
- Liu, G., Li, H., 2017. Offshore Platform Integration and Floatover Technology. Springer.
- Luo, H.B., Gao, Y., Wan, X.Z., Li, H.L., Wang, A.M., 2015. Field measurements of the sea environments and barge motions for the floatover installation of Kenli 3-2 mega topside in Bohai Bay. In: Proceedings of the 25th International Ocean and Polar Engineering Conference. Kona, Big Island, Hawaii, USA.
- Luo, M.Y.H., Edelson, D., Wang, J.N., Sun, J.L., Hassanaliaragh, S., 2013. Improvements in heavy topside installation onto spar hull by catamaran floatover method. In: Proceedings of the 32th International Conference on Offshore Mechanics and Arctic Engineering. Nantes, France.
- Ma, Y., Yuan, L., Zan, Y., Huang, F., 2018. Numerical simulation of float-over installation for offshore platform. *J. Mar. Sci. Appl.* 17 (1), 79–86.
- Magee, A., Yunos, N.Z.M., Kurian, V.J., 2014. Model testing capabilities for verification of floatover operations. In: Proceedings of the 24th International Offshore and Polar Engineering Conference. Busan, Korea.
- Maher, J.V., Prislis, I., Chao, J.C., Halkyard, J.E., Finn, L.D., 2001. Floatover Deck Installation for Spars. Offshore Technology Conference, Houston, Texas, USA.
- Mercier, J.A., et al., 1991. Tension leg platforms-progress and prospects. *SNAME Trans* 99, 249–279.
- Mouhandiz, A.A.E., Bokhorst, J., 2013. Analysis and offshore support for the float-over of a 24, 250mt topsides on the north west shelf. In: Proceedings of the 32th International Conference on Offshore Mechanics and Arctic Engineering. Nantes, France.
- ONEILL, L., Fakas, E., Ronalds, B., Christiansen, P., 2000. History, trends and evolution of float-over deck installation in open waters. In: SPE Annual Technical Conference and Exhibition (Dallas, Texas, USA).
- Paulling JR, 1995. MULTISIM Time Domain Platform Motion Simulation. Theoretical manual, third ed. Geyserville, California, November.
- Peng, R., Zhi, Z., 2012. A state-of-the-art review on structural health monitoring of deepwater floating platform. *Pac. Sci. Rev.* 14 (3), 253–263.
- Peters, D., Zimmer, R., Hein, J.W.M., Wang, W., Leverette, S., Bozeman, J., 1990. Weight Control, Performance Monitoring, and In-Situ Inspection of the TLWP. Offshore Technology Conference, Houston, Texas, USA.
- Ramzan, F.A., Rawstron, P.J.M., 1984. Installation techniques for integrated decks, the society of naval architects and marine engineers. In: International Maritime Innovation Symposium. New York, USA.
- Russell, G., 2017. Arkutun-dagi field development. In: 22nd World Petroleum Congress. Seij, M., Groot, H., 2007. State of the Art in Float-Overs. Offshore Technology Conference Houston, Texas, USA.
- Sigrist, J.H., Thomas, P.A., Naudin, J.C., 1998. Experience in Float over Integrated Deck - Flexibility of the Concept. Offshore Technology Conference, Houston, Texas, USA.
- Song, C.Y., Park, W.C., Lee, D.J., Lee, K.S., Sung, H.G., 2020. Design sensitivity of deck support frame structure for float-over installation. In: Proceedings of the 30th International Offshore and Polar Engineering Conference. Shanghai, China.
- Sun, L., Taylor, R.E., Choo, Y.S., 2012. Multi-body dynamic analysis of float-over installations. *Ocean. Eng.* 51, 1–15.
- Tahar, A., Halkyard, J., Steen, A., Finn, L., 2006. Float over installation method-comprehensive comparison between numerical and model test results. *J. Offshore Mech. Arctic Eng.* 128 (3), 256–262.
- Tao, W., Wang, S.Q., Song, X.C., Guo, Y.Z., Zhao, Y.X., 2020. A continuous simulation method of load transfer process during twin-barge float-over installation. In: Proceedings of the 30th International Offshore and Polar Engineering Conference. Shanghai, China.
- Tao, W., Wang, S.Q., Song, X.C., Li, H.J., 2021. Experimental investigation on continuous load transfer process of twin-barge float-over installation. *Ocean. Eng.* 234, 109309.
- Tan, B., Sahasrabudhe, S., Haney, J.A., Leow, B., Ray, J., 2008. Arthit Field Development: Float-Over Hardware Design and Issues. Offshore Technology Conference, Houston, Texas, USA.
- Tian, X., Wang, P., Li, X., Wu, X., Lu, W., Wu, C., Hu, Z., 2018. Design and application of a monitoring system for the floatover installation. *Ocean. Eng.* 150, 194–208.
- Tribout, C., Emery, D., Weber, P., Kaper, R., 2007. Float-overs Offshore West Africa, Offshore Technology Conference. Offshore Technology Conference, Houston, Texas, USA.
- Vernes, N., Donia, S., van Ginkel, A., de Jonge, J., Selstad, H., Krokeide, B., 2019. Aasta Hansteen Hull-Topsides Dry Transport and Dual Barge Mating Operations. Offshore Technology Conference, Houston, Texas, USA.
- Wang, A., Jiang, X., Yu, C., Zhu, S., Li, H., Wei, Y., 2010. Latest progress in floatover technologies for offshore installations and decommissioning. In: Proceedings of the 20th International Offshore and Polar Engineering Conference. Beijing, China.
- Wang, A.M., Jin, X.J., Liu, Y.Y., Tao, F.W., He, C., He, M., 2018. A low-deck floatover installation technology with strand jack lifting scheme. In: Proceedings of the 28th International Ocean and Polar Engineering Conference. Sapporo, Japan.
- Wang, A.M., Wang, T., Li, Y.C., Yu, C.S., Tao, F.W., He, C., 2014. A complete failure of a floatover installation with strand jack lifting scheme and a great success of float-on and float-off rescue operation. In: Proceedings of the 33th International Conference on Offshore Mechanics and Arctic Engineering. California, USA. San Francisco.
- Wang, P., Wu, X., Lu, W.Y., Tian, X.L., Li, X., 2018. A field monitoring system for dynamic positioning floatover installation. In: Proceedings of the 37th International Conference on Offshore Mechanics and Arctic Engineering. Madrid, Spain.
- White, G., Ramzan, F., Rawstron, P., Miller, B., 1986. Offshore Installation of an Integrated Deck onto a Preinstalled Jacket. Offshore Technology Conference, Houston, Texas. OTC-5260-MS.
- Wang, P., Tian, X., Peng, T., Luo, Y., 2017. A review of the state-of-the-art developments in the field monitoring of offshore structures, 2018 *Ocean. Eng.* 147, 148–164.
- Wu, X., Li, X., Li, X., Li, L., Yuan, S., Wang, A., 2014. Environmental and motion monitoring applications for challenging floatover installation of Liwan 3-1 mega topsides in South China Sea. In: Proceedings of the 24th International Offshore and Polar Engineering Conference. Busan, Korea.
- Xia, J., Hayne, S., Macfarlane, G., Field, D., Drobyshevskii, 2005. Investigation into floatover over installations of minimal platforms by hydrodynamic model testing. In: Proceedings of the 24 International Conference on Offshore Mechanics and Arctic Engineering. Halkidiki, Greece.
- Xu, N., Qin, L.C., Wang, Y.S., 2018. Application of dynamic positioning float-over technology to HZ25-8 DPP topside installation. In: Proceedings of the 28th International Ocean and Polar Engineering Conference. Sapporo, Japan.
- Xu, X., Li, X., Yang, J.M., Xiao, L.F., 2016. Hydrodynamic interactions of three barges in close proximity in a floatover installation. *China Ocean Eng.* 30 (3), 343–358.
- Xu, X., Yang, J.M., Li, X., Xu, L.Y., 2015. Time-domain simulation for coupled motions of three barges moored side-by-side in floatover operation. *China Ocean Eng.* 29 (2), 155–168.
- Yang, Y., Wang, A., Yuan, R., Li, H., Xie, W., Jin, X., 2014. Design considerations for challenging floatover installation of Liwan 3-1 mega topsides with a T-shaped barge. In: Proceedings of the 24th International Ocean and Polar Engineering Conference. Busan, Korea.
- Yi, F., Liu, T.F., Wang, L., 2018. Numerical simulation of float-over installation under the influence of fender with dynamically positioned vessel. In: Proceedings of the 27th International Ocean and Polar Engineering Conference. Sapporo, Japan.
- Yu, W.T., Li, H.L., Zhu, S.H., Wang, A.M., Xie, P., Luo, H.B., 2017. Comparison of field measurement and numerical simulation of the T-shaped barge motions during the topside floatover installation. In: Proceedings of the 36th International Conference on Ocean, Offshore and Arctic Engineering. Trondheim, Norway. Honolulu.
- Yuan, M., Chen, L., Li, Y., Liang, Y., Sun, X., Li, H.L., Xie, W.W., 2014. Design considerations of Liwan 3-1 topside float-over mating analysis. In: Proceedings of the 24th International Ocean and Polar Engineering Conference. Busan, Korea.
- Zhao, S.J., Meng, X., Li, H.J., Fu, Q., 2021. Continuous multi-body dynamic analysis of float-over deck installation with rapid load transfer technique in open waters. *Ocean. Eng.* 224, 108729.
- Zhu, L., Zou, M.Y., Chen, M.S., Li, L., 2021. Nonlinear dynamic analysis of float-over deck installation for a GBS platform based on a constant parameter time domain model. *Ocean. Eng.* 135, 109443.
- Zhu, S.H., Yu, W.T., He, M., Wang, A.M., Li, X., Wang, L., 2017. A comprehensive field measurement for floatover installations. In: Proceedings of the 27th International Offshore and Polar Engineering Conference. California, USA. San Francisco.
- Zu, Y., Chiu, C.Y., Cho, C.G., Wang, A., Ruskin, A., Qiu, X.H., 2016. DP conversion for a successful float-over installation. In: Proceedings of the 26th International Ocean and Polar Engineering Conference. Rhodes, Greece.